

HYDROGENATED NBR (HNBR)

SYNTHESIS AND PROCESS CONTROL:

PATENT LANDSCAPE REVIEW

1. ABSTRACT

This report synthesizes six patents focused on the preparation of hydrogenated nitrile rubber (HNBR) from nitrile rubber (NBR), with emphasis on selective hydrogenation, simultaneous hydrogenation/metathesis, and catalyst-mediated molecular-weight control. Across the landscape, the target material is consistently defined as an NBR-derived elastomer in which the butadiene-derived double bonds are partially to highly hydrogenated while nitrile functionality is retained, enabling improved heat, ozone, and oil resistance. The patents collectively address two major manufacturing challenges: achieving high residual-double-bond conversion without gelation or nitrile reduction, and reducing the molecular weight/Mooney viscosity of HNBR to improve processability.

Two broad process families dominate. The first is direct hydrogenation of NBR using homogeneous noble-metal catalysts, especially ruthenium, osmium, rhodium, or palladium systems, often in organic solvent and under elevated hydrogen pressure and temperature. US20180223005A1 and EP3827029A1 exemplify this route, with the former emphasizing catalyst composition design using Ru/Os complexes and vinyl-containing co-catalysts, and the latter focusing on catalyst residue management and aging stability through controlled noble-metal loading. The second family uses metathesis-assisted molecular-weight reduction, either as a pre-hydrogenation degradation step or as a simultaneous metathesis/hydrogenation process. US7919563B2, EP2473281B1, and US9079979B2 show that Ru/Os metathesis catalysts can be used with co-olefins and hydrogen to generate lower-Mw HNBR in a single step or staged sequence, while US8957161B2 provides supporting catalyst-system chemistry relevant to metathesis activation.

Key formulation/process trends include use of chlorobenzene or MCB-type aromatic solvents, hydrogen pressures from about 4.137 to 8.4 MPa in the examples, temperatures from 80 to 140°C, and catalyst loadings in the low-phr range or ppm metal range. Co-catalyst pretreatment with vinyl ethyl ether, use of 1-hexene as co-olefin, and careful control of catalyst identity strongly influence hydrogenation rate, molecular-weight retention, and final Mooney viscosity. Overall, the evidence supports Ru/Os-based catalyst systems and metathesis-enabled process designs as the most effective routes to low-Mw, processable HNBR.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20180223005A1
- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber using catalyst compositions formed by contacting a ruthenium- or osmium-based carbene complex with a vinyl-containing co-catalyst before hydrogenation. The process can also include prior molecular-weight degradation by metathesis, and the examples show hydrogenation of Perbunan® and NBR feedstocks in MCB under hydrogen pressure. - Catalyst-to-co-catalyst molar ratio: 1:(20-550) — Claim 1 - Catalyst-to-co-catalyst molar ratio: 1:(20-500) — Claim 3 - Catalyst-to-co-catalyst molar ratio: 1:(20 to below 100) — Claim 4 - Catalyst pretreatment temperature: -20 to 160 °C — Claim 2 - Hydrogenation temperature: 60 to 200 °C — Claim 5 - Hydrogen pressure: 0.5 to 35 MPa — Claim 5 - Example 1 hydrogenation temperature: 120 °C — Example 1/Table 2 - Example 1 hydrogen pressure: 4.137 MPa — Example 1/Table 2 - Example 1 agitation speed: 800 rpm — Example 1 - Example 1 hydrogenation time: 5 h — Example 1 - Example 1 catalyst amount: 15 mg Wilkinson's catalyst — Example 1 - Example 1 co-catalyst amount: 18 mg PPh₃ — Example 1 - Example 2 catalyst amount: 9 mg Catalyst (1) — Example 2 - Example 2 co-catalyst amount: 100 µL vinyl ethyl ether — Example 2 - Example 2 hydrogenation time: 3 h — Example 2 - Example 3 catalyst amount: 18 mg Catalyst (2) — Example 3 - Example 3 hydrogenation time: 1 h — Example 3 - Example 4 catalyst amount: 9 mg Catalyst (2) — Example 4 - Example 4 hydrogenation time: 2 h — Example 4 - Example 5 catalyst amount: 9 mg Catalyst (3) — Example 5 - Example 5 hydrogenation time: 3 h — Example 5 - Example 6 hydrogenation temperature: 140 °C — Example 6 - Example 6 hydrogen pressure: 8.4 MPa — Example 6 - Example 6 agitation speed: 600 rpm — Example 6 - Example 6 hydrogenation time: 2 h — Example 6/Table 3 - Example 6 1-hexene amount: 11.1 g — Example 6 - Example 6 catalyst amount: 135 mg Catalyst (2) — Example 6 - Example 7 hydrogenation temperature: 140 °C — Example 7 - Example 7 hydrogen pressure: 8.4 MPa — Example 7 - Example 7 agitation speed: 600 rpm — Example 7 - Example 7 hydrogenation time: 3 h — Example 7 - Example 7 catalyst amount: 135 mg Catalyst (2) — Example 7 - Example 7 co-catalyst amount: 0.75 g vinyl ethyl ether — Example 7 - Example 7 1-hexene amount: 11.1 g — Example 7 - Examples 8-10 hydrogenation temperature: 140 °C — Examples 8-10/Table 3 - Examples 8-10 hydrogen pressure: 8.4 MPa — Examples 8-10/Table 3 - Examples 8-10 hydrogenation time: 2 h — Examples 8-10/Table 3 - Examples 8-10 1-hexene amount: 4.0 phr — Examples 8-10/Table 3 - Example 11 hydrogenation temperature: 140 °C — Example 11 -

Example 11 hydrogen pressure: 8.4 MPa — Example 11 - Example 11 agitation speed: 600 rpm — Example 11 - Example 11 hydrogenation time: 4 h — Example 11 - Example 11 catalyst amount: 363 mg Catalyst (2) — Example 11 - Example 12 hydrogenation temperature: 140 °C — Example 12 - Example 12 hydrogen pressure: 8.4 MPa — Example 12 - Example 12 agitation speed: 600 rpm — Example 12 - Example 12 hydrogenation time: 4 h — Example 12 - Example 12 catalyst amount: 259 mg Catalyst (2) — Example 12 - Example 12 co-catalyst amount: 1.41 g vinyl ethyl ether — Example 12

Relevant Experimental Evidence - Example 1: Comparative hydrogenation with Wilkinson's catalyst reached 90.3% hydrogenation in 5 h at 120 °C and 4.137 MPa. - Example 2: Catalyst (1) with vinyl ethyl ether pretreatment improved hydrogenation to 93% in 3 h. - Example 3: Catalyst (2) with vinyl ethyl ether pretreatment achieved 99% hydrogenation in 1 h. - Example 6: NBR-5 underwent metathesis plus hydrogenation to 99.5% hydrogenation in 2 h at 140 °C and 8.4 MPa. - Example 12: Catalyst pretreatment with vinyl ethyl ether enabled >99% hydrogenation using lower catalyst loading than the comparison example.

Technical Relevance This patent is highly relevant because it directly addresses selective HNBR hydrogenation and shows how catalyst composition and co-catalyst pretreatment accelerate conversion while preserving polymer molecular weight. It provides practical operating windows for hydrogen pressure, temperature, and catalyst formulation in HNBR manufacture.

Patent 2

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions of nitrile rubber, using osmium- or ruthenium-based metathesis catalysts together with salts that accelerate the reaction. Although the examples are not HNBR-specific, the chemistry is directly relevant to NBR chain scission and molecular-weight reduction prior to hydrogenation. - Salt: LiBr — Example 7 - Salt amount: 1 mg LiBr — Example 7 - Salt: CsBr — Example 8 - Salt amount: 1 mg CsBr — Example 8 - Grubbs II catalyst amount: 10 mg — Example 7 - Chlorobenzene amount: 0.3 mL — Example 7 - CDCl₃ amount: 0.2 mL — Example 7 - Diethyl diallylmalonate amount: 0.15 mL — Example 7 - Room temperature: room temperature — Example 7 - Time: 2 min before substrate addition — Example 7 - Time: 30 min — Example 7 - Time: 60 min — Example 7 - Time: 15 min — Example 8 - Time: 30 min — Example 8 - Time: 60 min — Example 8 - Time: 90 min — Example 8 - Time: 150 min — Example 8

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis relative to the no-salt control. - Example 8: CsBr also accelerated metathesis, though less strongly than LiBr in the reported profile.

Technical Relevance This patent is relevant because it expands the metathesis catalyst toolbox used to degrade nitrile rubber before hydrogenation. Salt-accelerated metathesis can help lower molecular weight and improve the processability of the eventual HNBR.

Patent 3

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches preparation of low-molecular-weight HNBR by subjecting nitrile rubber to metathesis in the presence of Ru/Os catalysts and optionally a co-olefin, followed by hydrogenation. The process is designed to reduce molecular weight and narrow the molecular-weight distribution while retaining the HNBR backbone structure. - Target Mw: 30,000 to 250,000 — Claim 1 - Target Mooney viscosity: 3 to 50 — Claim 1 - Target PDI: less than 2.5 — Claim 1 - Conjugated diene content: 40% to 90% by weight — Claim 1 - Unsaturated nitrile content: 10% to 60% by weight — Claim 1 - Hydrogenation temperature: 80 °C — Table 1 - Co-olefin pressure: 500 psi — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Experiment 1 Mn: 73,000 — Table 2 - Experiment 1 Mw: 189,000 — Table 2 - Experiment 1 PDI: 2.59 — Table 2 - Experiment 1 Mooney viscosity: 43 — Table 2 - Experiment 2 Mn: 60,000 — Table 2 - Experiment 2 Mw: 136,000 — Table 2 - Experiment 2 PDI: 2.27 — Table 2 - Experiment 2 Mooney viscosity: 28 — Table 2 - Experiment 3 Mn: 31,000 — Table 2 - Experiment 3 Mw: 59,000 — Table 2 - Experiment 3 PDI: 1.90 — Table 2 - Experiment 3 Mooney viscosity: 3 — Table 2 - Experiment 4 Mn: 55,000 — Table 2 - Experiment 4 Mw: 111,000 — Table 2 - Experiment 4 PDI: 2.02 — Table 2 - Experiment 4 Mooney viscosity: 31 — Table 2 - Starting polymer Mn: 85,000 — Table 2 - Starting polymer Mw: 296,000 — Table 2 - Starting polymer PDI: 3.50 — Table 2 - Starting polymer Mooney viscosity: 73 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis/hydrogenation reduced Mw to 189,000 and Mooney viscosity to 43. - Example 2: Higher catalyst loading further reduced Mw to 136,000 and Mooney viscosity to 28. - Example 3: The lowest reported Mw and PDI were achieved, with Mw 59,000 and PDI 1.90. - Example 4: 1-Hexene as co-olefin produced a lower-Mw HNBR with Mw 111,000 and Mooney viscosity 31.

Technical Relevance This patent is directly relevant because it establishes a route to low-Mw HNBR by metathesis-assisted chain shortening before or during hydrogenation. It provides concrete property targets and demonstrates how co-olefin choice and catalyst loading influence final HNBR viscosity and dispersity.

Patent 4

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of partially unsaturated NBR in solution using ruthenium, palladium, or rhodium compounds, with emphasis on controlling catalyst residue to improve aging properties and product appearance. The process is framed around producing HNBR with reduced residual catalyst contamination rather than a specific metathesis-based molecular-weight reduction step. - Ruthenium content: 10 ppm to 200 ppm — Claim 1 - Ruthenium content: 10 ppm to 150 ppm — Claim 1 - Ruthenium content: 10 ppm to 120 ppm — Claim 1 - Ruthenium content: 10 ppm to 79 ppm — Claim 1 - Ruthenium content: 42 ppm to 79 ppm — Claim 1 - Palladium content: 20 ppm to <67 ppm — Claim 1 - Palladium content: 44 ppm to <67 ppm — Claim 1 - Rhodium content: 50 ppm to <270 ppm — Claim 1 - Rhodium content: 79 ppm to <270 ppm — Claim 1 - Rhodium content: 50 ppm to 250 ppm — Claim 1 - Rhodium content: 50 ppm to 170 ppm — Claim 1 - Rhodium content: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 °C — Claim 1 - Hydrogen pressure: 700000 to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Example catalyst amount: 0.45 parts magnesium-silicate-supported Pd catalyst — cited prior art in description - Ageing test: 4 days at 140 °C — description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it addresses a major practical issue in HNBR production: residual noble-metal contamination and its impact on aging and discoloration. It provides catalyst-loading and operating-condition windows that support cleaner HNBR manufacture.

Patent 5

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed

- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen using Ru/Os catalysts, optionally with a co-olefin, to produce low-molecular-weight HNBR in a single step. The process is specifically designed to lower molecular weight and narrow the molecular-weight distribution while maintaining hydrogenation efficiency. - Target Mw: 20,000 to 250,000 — Description - Target Mooney viscosity: 1 to 50 — Description - Target PDI: less than 2.8 — Description - Hydrogenation temperature: 140 °C — Example 1/Table 3 - Hydrogen pressure: 8.4 MPa — Example 1/Table 3 - 1-hexene amount: 4.0 phr — Table 3 - Example 1 catalyst amount: 0.041 phr Hoveyda catalyst — Table 1 - Example 1 metal content: 65.6 ppm — Table 1 - Example 2 catalyst amount: 0.045 phr Arlt catalyst — Table 1 - Example 2 metal content: 65.6 ppm — Table 1 - Example 3 catalyst amount: 0.060 phr Wilkinson's catalyst — Table 1 - Example 3 metal content: 65.6 ppm — Table 1 - Example 4 catalyst amount: 0.055 phr Grubbs II catalyst — Table 1 - Example 4 metal content: 65.6 ppm — Table 1 - Example 1 reaction time: 0 to 240 min — Table 2 - Example 1 RDB at 60 min: 2.5% — Table 2 - Example 1 RDB at 120 min: 0.6% — Table 2 - Example 1 RDB at 180 min: 0.5% — Table 2 - Example 1 RDB at 240 min: 0.4% — Table 2 - Example 2 RDB at 60 min: 7.0% — Table 2 - Example 2 RDB at 120 min: 4.5% — Table 2 - Example 2 RDB at 180 min: 3.9% — Table 2 - Example 2 RDB at 240 min: 3.5% — Table 2 - Example 3 RDB at 60 min: 15.4% — Table 2 - Example 3 RDB at 120 min: 3.1% — Table 2 - Example 3 RDB at 180 min: 1.3% — Table 2 - Example 3 RDB at 240 min: 0.7% — Table 2 - HNBR 1 Mw: 113371 g/mol — Table 3 - HNBR 1 Mn: 48352 g/mol — Table 3 - HNBR 1 PDI: 2.34 — Table 3 - HNBR 1 Mooney viscosity: 33.3 — Table 3 - HNBR 2 Mw: 133758 g/mol — Table 3 - HNBR 2 Mn: 54595 g/mol — Table 3 - HNBR 2 PDI: 2.45 — Table 3 - HNBR 2 Mooney viscosity: 41.3 — Table 3 - HNBR 3 Mw: 195304 g/mol — Table 3 - HNBR 3 Mn: 64130 g/mol — Table 3 - HNBR 3 PDI: 3.05 — Table 3 - HNBR 3 Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave the fastest hydrogenation profile and the lowest Mooney viscosity among the compared catalysts. - Example 2: Arlt catalyst also produced low-Mw HNBR, but with slower hydrogenation than Example 1. - Example 3: Wilkinson's catalyst showed slower hydrogenation and higher final Mooney viscosity than the inventive catalysts. - Example 4: Grubbs II catalyst produced HNBR with intermediate hydrogenation performance and lower Mooney viscosity than Wilkinson's catalyst.

Technical Relevance This patent is central to HNBR synthesis because it demonstrates a one-step simultaneous metathesis/hydrogenation route to low-Mw HNBR. It identifies catalyst classes and operating conditions that directly control residual unsaturation, molecular weight, and processability.

Patent 6

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in hydrogen using Ru/Os catalysts, optionally with a co-olefin, to produce low-molecular-weight HNBR. It emphasizes catalyst selection and compares hydrogenation performance across Grubbs III, Grubbs II, and Wilkinson-type systems. - Target Mw: 20,000 to 250,000 — Description - Target Mooney viscosity: 1 to 50 — Description - Target PDI: less than 2.8 — Description - Example 1 catalyst amount: 0.057 phr Grubbs III catalyst — Example 1 - Example 1 catalyst amount: 0.080 phr Grubbs III catalyst — Example 1 - Example 1 catalyst amount: 0.06 phr Wilkinson's catalyst — Example 1 - Example 1 catalyst amount: 0.055 phr Grubbs II catalyst — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 91.3 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 reaction time: 0 to 240 min — Example 1/Example 2 RDB at 60 min: 2.0% — Table 2 - Example 1 RDB at 120 min: 0.4% — Table 2 - Example 1 RDB at 180 min: 0.3% — Table 2 - Example 2 RDB at 60 min: 0.7% — Table 2 - Example 2 RDB at 120 min: 0.5% — Table 2 - Example 2 RDB at 180 min: 0.3% — Table 2 - Example 3 RDB at 60 min: 15.4% — Table 2 - Example 3 RDB at 120 min: 3.1% — Table 2 - Example 3 RDB at 180 min: 1.3% — Table 2 - Example 3 RDB at 240 min: 0.6% — Table 2 - Example 4 RDB at 60 min: 2.9% — Table 2 - Example 4 RDB at 120 min: 0.6% — Table 2 - Example 4 RDB at 180 min: 0.3% — Table 2 - HNBR 1 Mw: 191541 g/mol — Table 3 - HNBR 1 Mn: 68643 g/mol — Table 3 - HNBR 1 PDI: 2.79 — Table 3 - HNBR 1 Mooney viscosity: 38.2 — Table 3 - HNBR 2 Mw: 135772 g/mol — Table 3 - HNBR 2 Mn: 58259 g/mol — Table 3 - HNBR 2 PDI: 2.33 — Table 3 - HNBR 2 Mooney viscosity: 20.2 — Table 3 - HNBR 3 Mw: 195304 g/mol — Table 3 - HNBR 3 Mn: 64130 g/mol — Table 3 - HNBR 3 PDI: 3.05 — Table 3 - HNBR 3 Mooney viscosity: 69.3 — Table 3 - HNBR 4 Mw: 144212 g/mol — Table 3 - HNBR 4 Mn: 59851 g/mol — Table 3 - HNBR 4 PDI: 2.45 — Table 3 - HNBR 4 Mooney viscosity: 32.6 — Table 3

Relevant Experimental Evidence - Example 1: Grubbs III catalyst gave faster hydrogenation than Wilkinson's or Grubbs II at comparable metal content. - Example 2: A higher-loading Grubbs III catalyst produced the lowest Mooney viscosity reported in the table. - Example 3: Wilkinson's catalyst showed the slowest hydrogenation profile and the highest final Mooney viscosity. - Example 4: Grubbs II catalyst delivered intermediate hydrogenation performance and moderate Mooney viscosity.

Technical Relevance This patent is relevant because it shows that catalyst identity strongly affects hydrogenation rate and final HNBR properties in simultaneous metathesis/hydrogenation. It

supports the use of Ru-based catalysts to obtain lower-Mw, more processable HNBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US20180223005A1 and EP3827029A1 focus on direct hydrogenation of NBR in solution, whereas US7919563B2, EP2473281B1, and US9079979B2 emphasize metathesis-assisted molecular-weight reduction before or during hydrogenation.
- The metathesis-enabled patents use Ru/Os catalyst families and often include co-olefins such as 1-hexene or ethylene; US20180223005A1 instead relies on catalyst pretreatment with vinyl ethyl ether to tune hydrogenation activity.
- Operating temperatures overlap in the 120-140 °C range in the examples, but EP3827029A1 claims a broader 60-200 °C window and 0.7-15 MPa hydrogen pressure range, while the example-rich hydrogenation patents commonly use 4.137-8.4 MPa.
- US7919563B2 and EP2473281B1 explicitly target low molecular weight and low Mooney viscosity HNBR, with EP2473281B1 reporting Mw as low as 113,371 g/mol and Mooney viscosity 33.3, and US7919563B2 reporting Mw as low as 59,000 g/mol and Mooney viscosity 3.
- US9079979B2 shows that Grubbs III outperforms Grubbs II and Wilkinson's catalyst in early hydrogenation kinetics, while US20180223005A1 shows that catalyst pretreatment with vinyl ethyl ether can improve hydrogenation rate and preserve molecular weight.
- EP3827029A1 is distinct in prioritizing catalyst residue control and aging stability, highlighting ppm-level noble-metal windows rather than metathesis or chain-shortening chemistry.

4. CONCLUSION

The patent set shows that HNBR synthesis is best controlled by either highly selective direct hydrogenation with Ru/Os/Rh/Pd catalysts or by metathesis-assisted chain shortening followed by hydrogenation, with the latter route being most effective for low-Mw, low-Mooney products. The strongest evidence supports Ru-based catalyst systems, hydrogenation at roughly 120-140 °C, hydrogen pressures around 4-8.4 MPa in the examples, and co-catalyst/co-olefin strategies such as vinyl ethyl ether pretreatment or 1-hexene/ethylene addition to tune activity and molecular weight. For cleaner product and better aging, catalyst residue should be minimized, with EP3827029A1 indicating practical ppm windows for Ru, Pd, and Rh. Recommended synthesis parameters based on the evidence are: Ru/Os-based catalyst chemistry, aromatic solvent media such as MCB/chlorobenzene, hydrogenation near 140 °C, hydrogen pressure near 8.4 MPa where high conversion is needed, and metathesis-enabled molecular-weight reduction when low Mooney viscosity is the priority.

5. REFERENCES

- US20180223005A1 | Catalyst compositions and their use for hydrogenation of nitrile rubber | Not disclosed | US | 2018 | <https://patents.google.com/patent/US20180223005A1/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>